

The Effect of Collateralized Deposits on Banking Stability: Evidence from the 1918 Influenza Pandemic*

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Abstract

Following the banking turmoil of 2023, policymakers have proposed deposit insurance reforms to increase the stability of uninsured deposits. One proposal is to require the collateralization of large uninsured deposits. We examine the effectiveness of collateralization by analyzing the deposit stability of NY banks during the 1918 influenza pandemic, a period when economic uncertainty substantially increased and the FDIC was not yet established. NY state charter banks accepted regular and preferred deposits, where preferred deposits were secured by collateral and regular deposits were unsecured. Using a difference-in-differences approach, we estimate the behavior of preferred versus regular deposits during the 1918 pandemic, where identification comes from exogenous cross-sectional variation in the severity of the pandemic across New York. We find that the growth rate of preferred deposits increases in the severity of the pandemic whereas the growth rate of regular deposits decreases. These results suggest depositors value having their accounts collateralized in times of stress, suggesting that a policy requiring deposits not eligible for FDIC insurance to be collateralized, increases deposit stability.

Keywords: 1918 Influenza Pandemic, Deposit Stability, Deposit Insurance Reform

JEL classification: E44, G21, N22

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1 Introduction

The collapse of Silicon Valley Bank and other regional banks in early 2023 highlights the continued importance of deposit stability. The depositor runs on these banks both caused these banks to fail and also spurred the FDIC and Federal Reserve to invoke emergency provisions to stem concerns about a wider panic, which could lead to widespread runs.¹

In the aftermath of these bank failures, the FDIC released an overview of its insurance program (FDIC, 2023).² In the report, the FDIC outlines various ways to extend deposit insurance to cover a larger set of deposits, aiming to better meet its goal of promoting deposit stability, and thus maintaining confidence in the banking system. Another option considered by the FDIC is to require large deposits that are ineligible for FDIC insurance to be explicitly backed by securities, an arrangement called secured deposits. Given the collateral, these secured depositors should be less prone to withdrawing their deposits during moments of stress.

However, the effectiveness of pledging collateral as a means to reduce depositor run risk is not a settled question. Advocates assert that these secured deposits reduce depositor run risk and so limit a bank's reliance on uninsured deposits, which are known to be flighty.³ Critics, however, question whether depositors value collateral, especially in times of high uncertainty. Upon realizing that the bank may be undercapitalized, a secured depositor may prefer to withdraw its funds rather than incur the costs of participating in a resolution process and be exposed to the liquidity risks associated with selling the securities pledged as collateral in the marketplace. Indeed, immediately following the financial crisis of 2007-09, there was concern that the short-term secured funding that securities dealers relied upon was not stable during times of stress.⁴

This paper addresses this fundamental question about the effectiveness of pledging collateral on

¹The FDIC announced that even uninsured depositors would be fully covered at the failed Silicon Valley Bank and Signature Bank. The Federal Reserve created a liquidity facility, the Bank Term Funding Program, which provided funding based on the par value of securities (as opposed to the usual practice of using the market value of securities). Metrick (2024) provides a synopsis of these bank failures and the policy responses.

²For an overview of this report, see <https://www.fdic.gov/news/speeches/2023/spmay0123b.html>.

³For example, Martin et al. (2018) documents how uninsured deposits flee a distressed bank and Egan et al. (2017) demonstrate that banks where uninsured deposits make up a large share of total deposits can become unstable.

⁴For example, the Federal Reserve Banks of Boston and New York held a "Risks of Wholesale Funding" conference on August 13, 2014. However, more recent work by Copeland and Martin (2025) argues that short-term funding secured by Treasuries was quite stable over the 2007-09 financial crisis.

reducing depositor flightiness during times of stress. We study the NY banking system from 1914 to 1925, when banks offered both regular and preferred deposits. The former were the usual unsecured deposits, while the latter were secured by collateral. The FDIC had not yet been introduced, and so there was no federal-level insurance program, and New York did not offer a state-level deposit insurance program during this time period. The lack of deposit insurance highlights the potential value of collateral. Finally, in this period, the 1918 influenza pandemic created a large amount of economic uncertainty, which could have caused depositors to reassess the health of their banks and perhaps withdraw their money.

Preferred deposits were secured by government-issued securities (Treasuries as well as securities issued by state governments). This preference meant that in the event of a bank going bankrupt, valid and enforceable collateralization agreements would be honored, resulting in the repayment of preferred deposits before regular deposits. In contrast, regular depositors often regained access to their deposits only after those deposits had been discounted from their original face value. Offsetting the benefit of having their deposits protected by collateral, preferred deposits received an interest rate fixed by law at a level below the market rate received by regular deposits ([Morgan and Parker, 1915](#)).

Although the arrival of the 1918 pandemic is arguably exogenous, it coincides with the end of World War I. This makes the interpretation of coefficients estimated from time-series variation difficult, because alongside the introduction of the pandemic the U.S. economy was reorienting away from a wartime standing.⁵ Given this challenge, we rely upon cross-sectional variation. The severity of the 1918 pandemic across the U.S., as has been shown in the literature, appears to be randomly distributed. Leveraging this assumption, we study how differences in the severity of the pandemic affected the behavior of preferred depositors relative to regular depositors, as well as other items on banks' balance sheets.

For this study, we collect and construct a new data set that combines individual bank balance sheet data, bank location data, and county-level statistics on mortality due to the 1918 influenza epidemic. The data set is focused on New York, a state for which rich balance sheet information

⁵In an analysis of the impact of the 1918 pandemic on employment, [Bridgman and Greenaway-McGrevy \(2023\)](#) both illustrates and tackles this issue of confounding factors.

is available alongside county-level data on mortalities by cause of death. We first construct a data set on annual balance sheets of all state-chartered commercial banks and trust companies in NY from 1914 to 1925. These balance sheet data provide detailed information on both the assets and liabilities of banks.

We then merge in data on bank location and use this geographic information to link each bank to the relevant county-level influenza epidemic death rate. This county-level matching is meaningful because during this time unit-banking prevailed, meaning almost all banks operated a single branch. The balance sheet of each bank, then, was heavily influenced by local affairs and, in particular, by the severity of the pandemic in the county in which the bank operated. Combining these data sets allows us to exploit regional mortality variation to examine the relationship between the pandemic and the banks' balance sheets.

Our empirical approach is to use difference-in-differences to compare the growth rates of preferred deposits to the growth rates of regular deposits across banks with different levels of exposure to the 1918 pandemic. A host of bank-level controls as well as a rich set of fixed effects are employed to account for confounding factors. The results imply that regular and preferred depositors had opposing reactions to the severity of the pandemic. The results imply that moving from the 25th to the 75th percentile of our measure of pandemic severity results in a 0.42 percent decrease in regular deposit growth rates and a 0.39 percent increase in preferred deposit growth rates. Hence, during a time of stress, depositors value collateral, an effect which drives a 0.81 percent difference in growth rates between regular and secured deposits.

These results are quite robust. There is a concern that banks in cities, especially NYC, may have different business models than those in rural areas, and this mix may be driving the results. To address this issue, we re-estimate the regression using only those banks in financial centers and then, separately, using only banks outside of financial centers.⁶ In both cases, the same general result holds. In addition, to address concerns about serial correlation, we re-estimate the difference-in-differences regression on a short time sample and find similar results.

We then build upon these results by examining the impact of the pandemic on the asset side

⁶We defined the financial centers during this period of time as Albany, Buffalo, Rochester, and NYC.

of banks' balance sheets. In the same way that the uncertainty of the pandemic drove depositors towards safer products, do banks adjust their asset portfolios towards safer or more liquid assets? To answer this question, we use the fairly detailed data on banks' loans to construct two groups of loans. The first group, denoted Safer loans, are those that are collateralized by mortgages or securities. The second group, denoted Riskier loans, are those collateralized by other assets or not collateralized at all. The expectation is that banks with a larger exposure to the pandemic would shift their portfolio of loans towards the Safer group. To test this hypothesis, we use the same difference-in-differences method described above to compare the relative growth rates of these two loans groups across banks with different levels of exposure to the 1918 pandemic. The results are slightly less robust than those for depositors, but deliver the intuitive result that banks shift their loan portfolios towards Safer loans in the face of increasing severity of the pandemic. Both depositors and banks, then, reacted to the severity of the pandemic by looking for safety, either in terms of preferred deposits or loans backed by high quality collateral.

Although the analysis presented here is specific to New York, the results are informative more generally. First, NY banks were central to the U.S. banking system during this period, so understanding the stability of their deposit base during a time of stress is important for the U.S. banking system. Second, the NY banking system at this time was representative of the prevailing state banking systems throughout the U.S., so lessons learned from examining NY banks are generally applicable.

This paper is relevant to today's modern banking system given the continued debate on the size of the coverage of deposit insurance. The banking turmoil of 2023 revived concerns that individual banking failures could spread and develop into larger system-wide bank runs, even with the current deposit insurance program. In response, the FDIC proposed various ways to expand the coverage of deposit insurance, thereby reducing the amount of uninsured deposits and hopefully significantly lowering the probability of bank runs ([FDIC, 2023](#)). Although the costs of expanding deposit insurance coverage would depend on the details of plan, implementation will prove difficult due to the large volume of uninsured deposits. This challenge is especially pronounced in the current environment, as deposit insurance fund (DIF) balances are already below their required levels

following the failures of Silicon Valley Bank and Signature Bank in March 2023.⁷

Secured deposits provide an alternative way to increase deposit stability. Our results provide support that deposits backed by collateral are less flighty in times of significant economic stress compared to uninsured deposits. The costs to the bank of offering this service are not insignificant, as securities need to be posted as collateral and, in addition, there are variable costs associated with the ongoing valuation of the securities. An estimate of these costs, and how they compare to the costs of accepting uninsured deposits, are beyond the scope of this paper. However, an in-depth evaluation of the costs and benefits of secured deposits relative to uninsured deposits seems appropriate given the concerns today around the total value of uninsured deposits and potential instability this type of deposit brings to a bank's deposit base.

This paper contributes to the large literature on deposit stability. These studies emphasize that depositors become more sensitive to information when macroeconomic conditions worsen because they cannot distinguish the quality of banks. Theoretically, [Gorton \(1985\)](#), [Gorton and Pennacchi \(1990\)](#), [Chari and Jagannathan \(1988\)](#), [Calomiris and Kahn \(1991\)](#), and more recently [Eisenbach \(2017\)](#) and [Dang et al. \(2020\)](#), show how information affects bank funding stability. Empirically, [Park \(1991\)](#), [Saunders and Wilson \(1996\)](#), [Calomiris and Mason \(1997\)](#), and [Chen et al. \(2022\)](#) show that deposit flows are more sensitive to information about bank performance. In addition, [Anderson and Copeland \(2023\)](#) show that suppressing bank information helps banks manage deposit outflows during a time of financial stress. This paper contributes to the empirical literature by demonstrating that secured deposits, backed by less informationally sensitive collateral, are less flighty than unsecured deposits, which are more informationally sensitive and more dependent on a bank's financial health. Furthermore, the result that secured deposits increase in times of uncertainty relative to uninsured deposits supports the findings of [Martin et al. \(2018\)](#), which show insured deposits flowing into a distressed bank even as uninsured deposits flowed out.

This paper also adds to the literature examining the impact of the 1918 influenza epidemic,

⁷The Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 requires the FDIC to maintain \$1.35 in the fund for every \$100 of insured deposits. If the reserve ratio falls below 1.35 percent or is expected to fall within 6 months, the FDIC generally must adopt a restoration plan to restore the DIF reserve ratio to at least 1.35 percent within 8 years. As of the fourth quarter of 2024, the DIF balance stood at \$137.1 billion, compared to \$10,672.3 billion in insured deposits, resulting in a reserve ratio of 1.28 percent ([FDIC, 2025](#)).

in particular the strand which studies the relationship between pandemics and economic outcomes (Brainerd and Siegler, 2003; Barro, Ursúa and Weng, 2020; Correia, Luck and Verner, 2022; Jordà, Singh and Taylor, 2022; Bridgman and Greenaway-McGrevy, 2023).⁸ This paper stands apart from this existing literature because of our focus on the effect of a pandemic on financial stability, a question we are able to answer because of newly available micro-level balance sheet data during this period.

The remainder of this paper is organized as follows. Section 2 provides a historical background. Section 3 introduces the data and provides summary statistics. Section 4 describes the empirical specifications and presents the results. Section 5 discusses the policy implications for today's modern banking system and concludes.

2 Historical background

This section begins with a description of the NY banking environment during the 1910s and 20s and ends with a summary of the impact of the 1918 influenza epidemic in New York.

2.1 New York's banking system during the 1910s and 1920s

Throughout the early 1900s, New York State occupied a prominent and influential role within the banking industry. New York included all three tiers of the reserve pyramid—country, reserve, and central reserve banks—that operated during this period, a distinction shared by only two other states. The state was home to the metropolis of New York City, which served as the central money market of the U.S., medium-sized cities with active manufacturing and industrial bases, and many small towns in rural areas. Typically, country banks kept their reserves at larger institutions located in what became the reserve cities of Albany and Buffalo, as well as the central reserve city of New York City.⁹ Banks also maintained substantial deposits in industrial hubs such as Rochester to

⁸Other strands of the literature which focus on the 1918 influenza epidemic focus on wages, demographic composition, or mortality differences across socioeconomic classes (Noymer and Garenne, 2000; Mamelund, 2006; Garrett, 2009) or health outcomes (Karlsson, Nilsson and Pichler, 2014; Clay, Lewis and Severini, 2018; Keyfitz and Flieger, 1968; Almond, 2006).

⁹Mitchener and Richardson (2015) provides detailed information on the reserve system during this time.

facilitate the processing of payments.

This reserve system accentuates possible differences between country banks and those in financial centers. In particular, banks in financial centers were likely influenced by a broad range of factors, whereas country banks were primarily affected by local market conditions. Our benchmark results consider all banks in our sample, but in a robustness check, we redo the analysis for banks in financial centers and, separately, for country banks.

Over our sample period, New York's commercial banks accepted preferred and regular deposits. Preferred deposits were secured by U.S. government securities, but paid interest at a maximum rate of 2 percent. Regular deposits paid higher interest than preferred deposits, but they were not secured by collateral. As reference, during the 1920s, interest rates on demand deposits in banks in New York City ranged from one to four percent. Interest rates on time deposits averaged one to two percentage points higher.

The backing of preferred deposits by government securities was valuable as the nominal value of those bonds guaranteed eventual repayment of the nominal value of the deposits that they backed. Receivers of failed banks honored valid and enforceable collateralization agreements and tended to expedite repayment of preferred deposits. Hence, following a bankruptcy event, preferred depositors both were made whole and also were paid out before regular deposits.¹⁰

Most depositors could choose between depositing as preferred or regular depositors (just as most depositors could choose to make time or demand deposits). The law, however, required some depositors to only make preferred deposits. This group included custodians of funds, such as lawyers overseeing trusts; NY state chartered savings banks, savings and loans associations, credit unions, land banks, the government of the State of New York and its business entities, and municipal and county governments. In our sample period, 95% of preferred depositors came from these required groups. Corroborating the idea that preferred deposits offered the safety desired by some investors, bankers considered preferred deposits to be a stable source of funding ([Bankers' Magazine, 1909, p.69](#)).

A major change that happened during this time was the passage of the Federal Reserve Act

¹⁰Newspapers indicate that preferred repayments typically began within a month and finished within one year. Preferred depositors never lost funds deposited in banks in New York.

of 1913 and the establishment of the Federal Reserve. The Federal Reserve went into operation in November 1915 and national-charter banks were required to become members. State-charter banks, however, could choose whether to become members, and during our period of study take up of Federal Reserve System membership among NY state-charter banks was quite limited. Up through 1917 less than 40 state-charter banks choose to become members and, as such, we do not explore whether Federal Reserve System membership differentially affected state-charter bank behavior in the 1918 influenza pandemic. In Appendix Section [D](#) we provide more details on the take-up of Federal Reserve System membership by state-charter banks.

2.2 1918 Influenza Pandemic

An outbreak of influenza spread across the world in three waves over 1918 and 1919. This epidemic was severe, with about 500 million people infected globally (a quarter of the world’s population) and at least 50 million people killed. The 1918 pandemic did not spare the United States, where over half a million deaths are attributed to the epidemic ([Crosby, 1989](#)).¹¹

During the pandemic, monthly influenza and pneumonia death rates in New York spiked up from below 0.5 deaths per 1000 people to almost 5.71 deaths per 1000 (see Appendix Figure [A1](#)). Further, there was a large amount of variation in the severity of the pandemic across counties in New York, with some counties experiencing almost four times the mortality rate than others (see Figure [1](#) and the summary statistics in Appendix Table [A1](#)).

The drivers behind the variation in mortality rates across counties are not well understood. Although researchers have suggested possible explanations, including the demobilization of troops after WWI, there is little consensus on the underlying causes ([Report of the Committee on the Atmosphere and Man, 1923](#); [Crosby, 1989](#); [Brainerd and Siegler, 2003](#); [Kolata, 2011](#)). Examining the impact of the 1918 influenza across the U.S., [Brainerd and Siegler \(2003, p. 7-8\)](#) reports that the statistical evidence “supports the notion of influenza mortality as an exogenous shock to the population.” Furthermore, the mortality rates “appear to be randomly distributed and do not seem

¹¹ Similar to the COVID-19 virus, the 1918 influenza itself did not often cause death. Instead, secondary infections such as bacterial pneumonia, were typically the direct cause of death.

to be related to the level of economic development, climate or geography.” Based on these results from the existing literature, we consider the cross-county distribution of the 1918 influenza to be exogenous to our object of interest, which are the balance sheet variables of NY state-charter banks (see Section 4.2 for further details).

3 Data and summary statistics

To study how the influenza pandemic affected deposit flows and other balance sheet variables, we combine information on state-charter banks and trust companies with flu mortality statistics. To construct this novel dataset, we begin by collecting the balance sheets of all NY state-chartered banks and trust companies from the Annual Report of the Superintendent of Banks for the period 1914 to 1925.

The balance sheet data provide a substantial amount of detail on a bank’s assets and liabilities. Starting with liabilities, both preferred and regular deposits are reported along with other liabilities, such as short-term borrowings, equity, and deposits due to banks, trusts, and bankers (see Appendix Table B2 for more details). There is even more of a breakdown on the asset-side. For the purposes of this paper, we focus on securities holdings (which are almost all government securities) and loans. Banks report four types of loans, which, ranging from safest to riskiest are: (i) collateralized by securities, (ii) collateralized by houses, (iii) collateralized by other assets, and (iv) uncollateralized loans. We group the first two types of loans and label them “safer” loans based on the quality of the posted collateral. Accordingly, we group the last two types of loans and label them “riskier” loans.

Finally, these data also provide the location of each bank. This allows us to merge these data with county-level mortality statistics (discussed below), as well as determine whether the bank is in a financial center or not. During this time, the financial centers in NY were New York City, Albany, Buffalo, and Rochester. Banks outside of these locations are deemed to be rural banks. Our final data set includes a total of 373 state-chartered banks and trust companies which reported

December call information from 1914 to 1925.¹² Of these banks, 250 are rural and 123 are in financial centers. Of those in financial centers, 80 percent, or 99 banks, are located in NYC.

To these balance sheet data we add information on the severity of the pandemic based on county-level mortality rates due to influenza and pneumonia in New York.¹³ From the Annual Report of State Department of Health of New York, we collect data on deaths due to influenza and pneumonia along with population for each county in New York, from 1914 to 1925. Using these data, we then compute mortality rates, which are the ratio of deaths due to influenza and pneumonia over population.

To more cleanly capture the effect of the pandemic on mortality versus other factors, we develop and use an excess mortality rate.¹⁴ This excess rate is equal to the mortality rate in the data minus a predicted rate, which is derived from a linear regression model that does not account for the arrival of the 1918 influenza pandemic.

Using data from 1914 to 1925, with 1918 and 1919 excluded, we regress the ratio of deaths from influenza and pneumonia over population in county i in year t on the natural log of county i 's population (denoted $\ln \text{pop}_{it}$) with county-level fixed effects. Formally, the specification is:

$$r_{i,t} = \alpha_0 + \alpha_1 \ln \text{pop}_{i,t} + \eta_i + \varepsilon_{i,t}, \quad (1)$$

where η denote county fixed effects and ε is a error term. Details of this regression are reported in Appendix C.

We then construct a measure of excess mortality rate, denoted $x_{i,t}$, for each county in 1918 and 1919 using the difference between the predicted values of the ratio of deaths to population, denoted

¹²We do not include data on national-charter banks for two reasons. First, those data are reported annually in September, a reporting cycle that does not align well with capturing the effects of the 1918 pandemic, which struck NY with intensity starting in October 1918 (see Appendix Figure A2). Second, the data collected on national-charter banks at this time was not detailed enough for the analysis presented here. In particular, the data do not distinguish between regular and preferred deposits nor break down bank loans with the same granularity.

¹³Use of mortality rates to measure severity of the pandemic is common in the literature, as in Garrett (2009); Barro, Ursúa and Weng (2020).

¹⁴Previous studies (Karlsson, Nilsson and Pichler (2014); Basco, Domènech and Rosés (2021)) also use excess death rates to study the impact of the 1918 Influenza Pandemic on economic performance.

$\hat{r}_{i,t}$, and the actual values,

$$x_{i,t} = r_{i,t} - \hat{r}_{i,t}. \quad (2)$$

Importantly, the predicted mortality rates do not account for the arrival of the 1918 pandemic, but rather projects what the rate would have been without the pandemic.¹⁵

We merge this county-level information on excess mortality rates with the balance sheet data, using the geographic information of the bank's branch available from the Annual Report of the Superintendent of Banks. This enables us to link detailed pandemic-related outcomes at the county level to individual bank-level balance sheet variables.

As an overview of how the pandemic effected banks, in Table 1 we report balance sheet ratios over the sample period, from 1914 to 1925. These ratios are further reported for three sub-periods: Pre-pandemic (1914-1917), Pandemic (1918-1919), and Post-pandemic (1920-1925). We start by examining the portfolio of assets held by banks, where all variables are normalized by total assets, to allow for a more meaningful comparison across banks of different sizes. We find that Securities and Total Loans are the largest asset categories, accounting for three-tenths and one-half of total assets, respectively (see Table 1). The most liquid assets held by banks are vault and interbank reserves. We label the sum of both types as reserves and find this measure accounts for about 19 percent of total assets. Finally, we provide statistics on the composition of loans held on banks balance sheets by considering the aforementioned safer and riskier loan types.

The change in ratios for loans and securities holdings present an intuitive pattern. With the onset of the pandemic, the average ratio of loans to assets declines from 53.7 percent to 49.8 percent whereas the average ratio of securities to assets increases from 24.1 percent to 30.3 percent. Such an outcome is consistent with banks shifting their portfolio of assets towards more liquid assets in response to the uncertainty of the pandemic. Alternatively, demand for loans could have fallen.

Looking at the portfolio of loans, we find that riskier loans dominate with 82% share of all

¹⁵A comparison of the mortality rate and excess mortality rates in the pandemic period is provided in Appendix Table A1, which reports summary statistics for both variables.

loans over the sample period.¹⁶ Somewhat counterintuitively, the share of riskier loans increases in the pandemic, from 81.4% to 84.1%, before falling back off to 81.7% in the post-pandemic period.

Turning next to the capital structure of banks, we consider equity capital and deposits as shares of total liabilities. Not surprisingly, deposits make up the lion's share of liabilities at 82.5%, with regular deposits making up over nine-tenths of deposits. The table illustrates that regular deposits slightly decreased in the pandemic, falling from 93.1% of total deposits in the pre-pandemic period to 92.8%. Regular deposits then recovered in the post-pandemic period, rising to 95.3% of total deposits. By construction, preferred deposits followed the opposite pattern.

Table 1 presents the affects of the pandemic on bank balance sheets in the time-series. As mentioned previously, because the pandemic overlapped with other major macroeconomic events, such as the end of WWI, the interpretation of the time-series variation is not straightforward. The constructed dataset, however, allows for a cross-sectional analysis, as the severity of the pandemic varies tremendously across NY counties. Given this severity is random, and so exogenous to the variables of interest, this cross-sectional variation allows for a clean interpretation of the estimated coefficients as capturing the effect of the pandemic, separate from confounding macroeconomic forces.

In the next section, we rigorously explore how this cross-sectional variation affected banks liabilities and assets. Below, however, we provide the intuition for the results, using binned scatter plots. We start with bank liabilities and focus on the variation of the growth rates of regular and preferred deposits with the severity of the pandemic. Figure 2 illustrates how regular deposit growth rates are negatively correlated with the severity of the pandemic. The estimate from a best-fit line imply that a one standard deviation increase in severity (an increase of 2.09 in the excess mortality rate) is associated with a 0.11 decrease in the growth rate of regular deposits. In stark contrast, preferred deposit growth rates are positively correlated. The best-fit line estimate shows a stronger reaction—a one standard deviation increase in severity predicts a 0.39 increase in the

¹⁶Mortgages during our sample period were typically short-term, interest-only loans with 2-5 years of maturity and were mostly farm mortgages, which were not very common in the Mid-Atlantic region. The boom of “originate and sell” (the most common mortgage business model today) started around the middle of the 1900s. See Snowden (2013) for more details.

preferred deposit growth rate.

We then turn to bank assets and focus on the composition of bank loans. We consider how the growth rates of safer and riskier loans vary with pandemic severity, once again using a binned scatter plot. The results are illustrated in Figure 3 and show that the growth rates of both types of loans decrease in the severity of the pandemic. The estimates from the best fit lines predict that a one standard deviation increase in severity results in a decrease of 0.04 in the growth rate of safer loans and a decrease of 0.10 in the growth rate of riskier loans. This suggests that banks, like depositors, reacted to the 1918 pandemic by shifting their portfolio of assets towards those with a safer profile. With this intuition, we now turn to a formal analysis of the data.

4 Empirical analysis

In this section, we present the empirical approach and results. We first describe the difference-in-differences approach and how the coefficients of interest are identified. We then discuss the estimated coefficients.

4.1 The difference-in-differences approach and identification

We start by examining the behavior of regular versus preferred deposits during the 1918 influenza pandemic. We then supplement this analysis with an examination of banks' loan portfolio of assets over the same period. For both, we use a difference-in-differences approach that combines depositor-level (asset-level) responses with cross-county differences in the severity of the pandemic. The key identifying assumption is that both the timing of the outbreak in NY and the differences in exposure across the state represent an exogenous shock. Further, depositor (bank) behavior would have been similar across counties in the absence of the pandemic.

The statistical analysis is a classic fixed-effects linear regression relating an indicator for the pandemic and a measure of the severity of the pandemic to various aspects of bank behavior, conditional on a set of bank-level controls and a rich set of fixed effects. The first specification

focuses on explaining the differential growth rates of regular versus preferred deposits.¹⁷ Let $D_{i,j,t}$ denote the level of deposits of type j at bank i in year t , where deposit type is preferred or regular. The dependent variable is the log difference in deposits, or $\Delta D_{i,j,t} = \ln D_{i,j,t} - \ln D_{i,j,t-1}$. The timing of the pandemic is captured with the indicator variable P_t , which is equal to 1 for the years 1918 and 1919. The severity of the pandemic across counties is captured by $x_{i,t}$, the excess mortality rate in year t in the county in which bank i is located. Finally we define the indicator variable PD_j to take on the value of 1 for preferred deposits and 0 otherwise, which allows for a differential response across the two types of deposits. The benchmark specification is

$$\Delta D_{i,j,t} = \beta_1 PD_j + \beta_2 P_t + \beta_3 PD_j \times P_t + \beta_4 P_t \times x_{i,t} + \beta_5 PD_j \times P_t \times x_{i,t} + \Gamma' B_{i,t-1} + \tau_t + v_{i,j} + \epsilon_{i,j,t}, \quad (3)$$

where the coefficients of interest are $\{\beta_4, \beta_5\}$, because they estimate how the severity of the pandemic differentially affected regular and preferred deposits. $B_{i,t-1}$ is a vector of bank-level controls, including liquid assets over total assets, equity capital over total liabilities, a measure of loan quality (amount secured over total loans), the log of total assets, and the log of the age of the bank (measured in years). We lag these bank controls to reflect the fact that the public receive balance-sheet information with a lag and to reduce potential endogeneity concerns. To control for unobserved time-varying economic factors, there is a set of year fixed effects, τ_t . Further, to control for heterogeneity at the bank level, there is a set of bank fixed effects, $v_{i,j}$, for each depositor type. The latter fixed effects allow us to control for unobserved heterogeneity at the (bank $\times$ depositor-type) level. The random error term is $\epsilon_{i,j,t}$, which is a mean-zero, possibly heteroskedastic and autocorrelated within-bank error term.

Supplementing the analysis focused on deposits, we examine the evolution of the portfolio of loans held by banks. Let $L_{i,j,t}$ denote the total amount of loans of type j held by bank i on year t , where type j distinguishes between safer and riskier loans (as defined in Section 3). The focus remains on growth rates, and so the dependent variable is $\Delta L_{i,j,t} = \ln L_{i,j,t} - \ln L_{i,j,t-1}$. Define the

¹⁷This approach draws from [Karas et al. \(2013\)](#).

indicator variable LR_j to take on the value of 1 for riskier loans and 0 otherwise, which allows for a differential response across the two types of loans. The specification, mirroring that in equation (3), is

$$\Delta L_{i,j,t} = \alpha_1 LR_j + \alpha_2 P_t + \alpha_3 LR_j \times P_t + \alpha_4 P_t \times x_{i,t} + \alpha_5 LR_j \times P_t \times x_{i,t} + \Gamma' B_{i,t-1} + \tau_t + \nu_{i,j} + \varepsilon_{i,j,t}. \quad (4)$$

In both (3) and (4), the coefficients of interest are those related to the cross-sectional variation across counties in the severity of the pandemic. In (3), these coefficients are $\{\beta_4, \beta_5\}$, and they capture the behavior of depositors. In (4), the coefficients of interest are $\{\alpha_4, \alpha_5\}$ and they capture how banks are managing their loan portfolios. These coefficients are identified because the impact of the 1918 influenza across New York is arbitrary and random, as characterized by the existing literature (see details in Section 2.2). As such, it is reasonable to consider the spread and severity of the 1918 influenza as exogenous to deposit growth and the growth rate of banks' loans.

In both specifications, an overall effect of the pandemic on the dependent variable is also estimated. But the interpretation of this effect is murkier because other macroeconomic effects occurred coincidentally. In particular, the end of WWI occurred on November 11, 1918 and resulted in large changes to the economy through the return of soldiers to civilian life and the restructuring of the economy towards a peacetime setting. As a result, the estimated overall effects of the pandemic should be treated with caution and, as such, are not emphasized in this analysis. In contrast, the estimated effects in reaction to the severity of the pandemic are cleanly interpretable.

4.2 Checking the exogeneity of the 1918 pandemic to banking activity

The existing literature argues that the spread and severity of the pandemic across the U.S. is not related to economic conditions (e.g., see [Brainerd and Siegler \(2003\)](#)). Nevertheless, as a check on this exogeneity assumption, we provide summary statistics on our objects of interest (growth rates of various balance sheet variables) in the pre-pandemic period, based on pandemic-related mortality rates. Specifically, we categorize banks based on the quantiles of the influenza mortality-

rate distribution in the pandemic period. Hence, banks in counties with the highest mortality rates in the pandemic are in the top category, whereas banks in counties with the lowest mortality rates are in the bottom category.

These summary statistics are reported in Table 2 and demonstrate that in the pre-pandemic period, there were no systematic differences across banks with different exposures to the pandemic; banks in each mortality-rate group were behaving similarly in terms of growth rates. For example, preferred deposit growth in the pre-pandemic period averaged 0.219 for banks in the 20-40 quintile, a rate higher than banks in the 60-80 quintile but lower than those in the 80-100 quintile. Furthermore, any difference in the mean growth rate across groups is quite small compared to the magnitude of the standard deviation of these growth rates. As such, we argue there were no systematic differences across banks located in counties that had different exposures to the 1918 influenza pandemic.

4.3 Empirical results on deposits

We begin with the results focused on the comparison of preferred versus regular deposits. We estimate equation (3) for all the banks in our sample. We then split the sample based on whether banks were in financial centers (i.e., NYC, Albany, Buffalo, and Rochester) or not and re-estimate the specification for each group of banks. Standard errors are clustered by bank and deposit type.¹⁸ The results are collected in Table 3.

Our main focus is on the differential behavior of deposit types in the pandemic, in response to the severity of the pandemic. The results imply that during the pandemic, regular deposits in the more affected areas experienced larger deposit outflows (see the negative coefficient on *Pandemic* $\times$ *Excess death*), but preferred deposits in these areas saw deposit inflows (note the sum of the coefficients on *Pandemic* $\times$ *Excess death* and *Preferred* $\times$ *Pandemic* $\times$ *Excess death* is equal to 0.114). The change in deposits is meaningful; moving from the 25th to 75th percentile of pandemic

¹⁸We follow the general principle of clustering at the most granular level possible (Cameron and Miller, 2015), while accounting for how error terms are correlated, such as autocorrelations at the bank level and the different distributions of error terms across deposit types. For example, regular deposits might be more randomly dispersed compared to preferred deposits (Abadie et al., 2023).

severity results in a 0.42 percent decrease in regular deposit growth rates.¹⁹ At the same time, there is a 0.39 percent increase in preferred deposit growth rates. Therefore, in total, there is a 0.81 percentage point difference in growth rates between preferred and regular deposits. In Figure 4, the marginal effects of pandemic severity on deposit growth rates are illustrated. The marginal effects are quite a bit larger when moving from the 50th to 75th percentiles compared to moving from the 25th to 50th percentiles, reflecting the fact that the 1918 pandemic had an uneven impact across NY, with some counties suffering from especially high mortality rates. Although similar in magnitude, the changes in regular deposits have a much larger impact on a bank's balance sheet given that regular deposits are, on average, about 78 percent of total liabilities.²⁰

Given that depositors' responses to economic uncertainty may have differed depending upon banks business models (see background discussion in Section 2.1), we study banks in financial centers and rural areas separately. The same general effect is found across both groups of banks although there are differences in magnitude. Looking at banks in financial centers first, the results imply a larger decline in regular deposits given an increase in pandemic severity, compared to the estimates including all banks (compare -0.380 to -0.121 in the first row of Table 3). Indeed, moving from the 25th to 75th percentile in pandemic severity results in a decrease of 1.30 percent in regular deposits for banks in a financial center. Further, the estimated coefficients imply that preferred deposits do not increase in pandemic severity: The linear combination of the two relevant coefficient estimates is equal to -0.038 and has a standard error of 0.126.²¹ Figure 5 illustrates these marginal effects for banks in financial centers.

The estimated results for rural banks mirror those for all banks. An increase in pandemic severity causes regular deposits to decrease and preferred deposits to increase. These marginal effects are illustrated in Figure 5 and show that for rural banks, the marginal effect is larger for preferred deposits.

As a robustness exercise, we re-estimate this regression over a shorter sample period of 1916-

¹⁹As reported in Appendix Table A1, the interquartile range for excess mortality rates is 3.43.

²⁰As noted in Table 1, in our sample, regular deposits account for 94.2 percent of total deposits, and total deposits account for 82.6 percent of total liabilities.

²¹The linear combination of *Pandemic* $\times$ *Excess death* and *Preferred* $\times$ *Pandemic* $\times$ *Excess death* is highly statistically significant for both the all banks and rural banks cases.

1919. This approach of having only two observations pre-pandemic and two observations in the pandemic addresses concerns about serial correlation.²² The results are presented in Appendix Tables E4 and demonstrate that the results are robust to this potential criticism. Indeed, for the all banks case, the estimated coefficients on $Pandemic \times Excess\ death$ and $Preferred \times Pandemic \times Excess\ death$ are only slightly different.

The above analysis compares regular to preferred deposits. In addition to trusts and other sophisticated investors, another significant holder of preferred deposits are state governments. Given the different nature of state governments, we explore whether this type of depositor acted in the same way as other preferred depositors. We accomplish this by re-estimating equation (3) where preferred deposits is replaced with the preferred deposits held by state governments. Once again, we estimate this specification for all banks, those banks in financial centers, and rural banks.

The results are collected in Table 4 and the estimated coefficients are quite similar to those estimated in the benchmark case. These results imply that state governments act similarly to other types of preferred depositors, in that they increase their preferred deposits in reaction to increases in the severity of the pandemic.

4.4 Empirical results on loans

We then turn to the results on loans, focused on the comparison of safer versus riskier loans over the pandemic. Mirroring the analysis of deposits, we estimate equation (4) for all banks in the sample, and then consider the sub-samples of banks in financial centers and those that are not. The results are collected in Table 5.

For the sample of all banks and financial center banks, the estimated coefficients imply a difference in growth rates across these two loan types given changes in the severity of the pandemic. The coefficient on the interaction term of riskier loan growth, the pandemic indicator, and the severity of pandemic is negative, and statistically significant in both of these specifications. In the sample of all banks, the estimated combined effect of a change in pandemic severity, however, is -0.006,

²²The dangers of serial correlation in typical difference-in-differences exercises are illuminated in [Bertrand et al. \(2004\)](#). That paper proposes the robustness exercise of collapsing the data so as to rely on cross-sectional variation, which has been adopted here.

an economically small effect (and also statistically insignificant).²³ For banks in financial centers, the linear combination of the two estimated coefficients is a larger -0.099, however the standard error is 0.073, implying this effect is not statistically different from zero.

A statistically and economically significant effect is found when considering the specifications estimated on a shorter sample period of 1916 to 1919. Following the approach taken with deposits, we re-estimate the regressions on this shorter sample period as a robustness exercise. The estimated coefficients are presented in Appendix Table E5. In the shorter sample, the estimated coefficients imply that safer loans have greater growth in the severity of the pandemic alongside a decline in the riskier loan growth. For the all-bank and rural-bank specifications, the increase in safer loan growth rate is statistically and economically meaningful. Moving from the 25th to 75th quartile of pandemic severity increases the growth rate of safer loans by 0.12 (0.11) percent in the all bank (rural bank) specification. The results also imply that the effect of a change in pandemic severity does not have a statistically significant effect on riskier loan growth.

Taken together, the results on the full and shorter sample periods imply that banks shift their loan portfolio towards holding safer assets given an increase in the severity of the pandemic. These results accord with those on deposits, in that just as the uncertainty of the pandemic drove depositors towards safer products, it also pushed banks to adjusted their asset portfolios towards loans collateralized by higher quality assets.

5 Conclusion

This paper studies whether deposits secured by collateral are less flighty than uninsured deposits during times of stress. This fundamental question is addressed by examining the behavior of preferred and regular deposits of state-chartered banks in NY during the 1918 influenza pandemic, when depositors faced elevated economic uncertainty. Regular deposits are uninsured and so pose the usual run-risks to banks during times of stress. In contrast, preferred deposits are backed by government-issued securities, and so potentially are more stable. We use a difference-in-

²³The standard error of the linear combination of the two interaction terms is 0.011.

differences approach and rely upon cross-sectional variation in the severity of the pandemic across NY to identify the reaction of each type of deposit to the elevated economic uncertainty. We find that regular and preferred deposits had opposite reactions. Regular deposit growth decreased in the severity of the pandemic whereas preferred deposit growth increased. An increase in the severity of the pandemic from the 25th to 75th percentile results in a 0.81 percent difference in growth rates between regular and preferred deposits.

This study has important implications for policy today. Following the banking turmoil of early 2023 and the concerns that the failures of a few banks could spur a larger run on banks, policymakers issued proposals to stabilize the deposit base. In particular, in May 2023, the FDIC proposed several options to enhance deposit stability, with detailed information on the advantages and disadvantages of each option and what additional policies can be implemented simultaneously to best balance the risk of moral hazard. One of the proposed options is to require secured deposits for large uninsured deposits.

However, policymakers question whether posting securities as collateral for a deposit would discourage the risk of bank runs by these depositors during times of stress. Advocates assert that such requirements can limit banks' reliance on uninsured deposits, reduce depositor run incentives, and increase depositor discipline. Critics have questioned its ability to stop bank runs, especially if an institution is considered insolvent. Upon realizing that a bank may be undercapitalized, a secured depositor could prefer withdrawal to recouping collateral in a resolution process and becoming exposed to the liquidity risk inherent to selling the securities in the market place.

Our study shows that secured deposits contribute to the stability of a bank's deposit base. Our results imply secured deposits should be considered as an another way to increase deposit stability alongside the more traditional proposals to expand the set of deposits eligible for the FDIC's insurance program.

Table 1: Balance Sheet Ratios, by Period

	All Periods	Pre-Pandemic	Pandemic	Post-Pandemic
Liquid assets	0.193 (0.116)	0.247 (0.125)	0.204 (0.122)	0.158 (0.094)
Securities	0.281 (0.156)	0.241 (0.150)	0.303 (0.146)	0.297 (0.158)
Loans	0.536 (0.148)	0.537 (0.144)	0.498 (0.134)	0.547 (0.152)
Safer loans	0.180 (0.147)	0.186 (0.155)	0.159 (0.131)	0.183 (0.146)
Riskier loans	0.820 (0.147)	0.814 (0.155)	0.841 (0.131)	0.817 (0.146)
Equity	0.134 (0.075)	0.153 (0.069)	0.121 (0.060)	0.127 (0.080)
Deposits	0.825 (0.090)	0.812 (0.078)	0.814 (0.089)	0.836 (0.096)
Regular deposits	0.942 (0.089)	0.931 (0.103)	0.928 (0.076)	0.953 (0.082)
Preferred deposits	0.058 (0.089)	0.069 (0.103)	0.072 (0.076)	0.047 (0.082)

Note: Cell entries are means, with standard deviations in parenthesis. The Pre-Pandemic period is from 1914 to 1917. The Pandemic period is 1918 through 1919. The Post-Pandemic period is the years from 1920 through 1925. Liquid assets, securities, loans, equity and deposits are reported as shares of total assets. Safer and riskier loans are reported as shares of total loans and regular and preferred deposits are reported as shares of total deposits.

Source: *Annual Report of the Superintendent of Banks* and authors' calculations.

Table 2: Mean and Standard Deviation of Balance Sheet Growth Rates for 1914-1917, by Influenza Mortality Severity Quintile in 1918

	(0-20)	(20-40)	(40-60)	(60-80)	(80-100)	All
Liquid assets	0.192 (0.355)	0.215 (0.352)	0.172 (0.328)	0.214 (0.304)	0.092 (0.378)	0.181 (0.337)
Securities	0.150 (0.372)	0.209 (0.400)	0.204 (0.431)	0.253 (0.637)	0.229 (0.309)	0.210 (0.459)
Loans	0.093 (0.176)	0.061 (0.140)	0.097 (0.183)	0.088 (0.142)	0.067 (0.143)	0.087 (0.166)
Safer loans	0.063 (0.283)	0.047 (0.234)	0.002 (0.442)	0.089 (0.270)	0.144 (0.331)	0.049 (0.358)
Riskier loans	0.096 (0.199)	0.070 (0.192)	0.107 (0.232)	0.091 (0.178)	0.062 (0.167)	0.093 (0.207)
Equity	0.050 (0.145)	0.047 (0.085)	0.034 (0.103)	0.048 (0.109)	0.041 (0.080)	0.041 (0.106)
Deposits	0.142 (0.150)	0.148 (0.127)	0.151 (0.176)	0.159 (0.127)	0.133 (0.129)	0.149 (0.153)
Regular deposits	0.137 (0.144)	0.141 (0.130)	0.144 (0.173)	0.157 (0.126)	0.131 (0.129)	0.144 (0.151)
Preferred deposits	0.157 (0.719)	0.219 (0.476)	0.202 (0.604)	0.195 (0.564)	0.283 (0.748)	0.205 (0.609)

Note: Columns are groups of banks, categorized by the severity of influenza mortality rates in 1918. The first column includes banks located in counties that fall into the lowest quintile of 1918 mortality rates. The second column are those banks located in counties that fall into the second lowest quintile mortality rates, and so on. Cell entries are means, with standard deviations in parentheses, over the 1914 to 1917 period. Liquid assets, securities, loans, equity and deposits are reported as shares of total assets. Safer and riskier loans are reported as shares of total loans and regular and preferred deposits are reported as shares of total deposits.

Source: *Annual Report of the Superintendent of Banks* and authors' calculations.

Table 3: Depositors' Reactions to the Severity of Pandemic, 1914 to 1925

	All	Financial Center	Rural
Pandemic x Excess Death	-0.121*** [0.024]	-0.380*** [0.137]	-0.101*** [0.023]
Preferred x Pandemic x Excess Death	0.235*** [0.025]	0.342*** [0.055]	0.200*** [0.027]
Total assets (t-1)	-0.307*** [0.043]	-0.306*** [0.065]	-0.316*** [0.054]
Reserves to assets (t-1)	-0.002 [0.002]	0.002 [0.003]	-0.004* [0.002]
Equity to liabilities (t-1)	0.000*** [0.000]	0.000*** [0.000]	-0.000 [0.000]
Loan quality (t-1)	-0.000 [0.001]	-0.001 [0.001]	0.000 [0.001]
Bank age (t-1)	0.076 [0.051]	0.237** [0.094]	-0.015 [0.054]
Year Fixed Effects	Yes	Yes	Yes
Bank x Deposit-Type Fixed Effects	Yes	Yes	Yes
Observations	5,297	1,594	3,703
R-squared	0.148	0.193	0.140

Note: Dependent variable is deposit growth, measured as the first difference of the natural logarithm of bank deposits. Pandemic is an indicator variable equal to 1 in 1918 through 1919, Preferred is an indicator variable equal to 1 for preferred deposits, and Excess Death is an excess mortality-based measure of the impact of the pandemic at the county level. Standard errors are clustered by bank and deposit type, and are reported in parenthesis.

* denotes a p-value less than 0.10, ** denotes a p-value less than 0.05, *** denotes a p-value less than 0.01.

Table 4: Depositors' Reactions to the Severity of Pandemic, State Deposits, 1914 to 1925

	All	Financial Center	Rural
Pandemic x Excess Death	-0.159*** [0.042]	-0.493*** [0.176]	-0.134*** [0.043]
Preferred x Pandemic x Excess Death	0.269*** [0.043]	0.410*** [0.071]	0.214*** [0.054]
Total assets (t-1)	-0.389*** [0.055]	-0.360*** [0.077]	-0.416*** [0.078]
Reserves to assets (t-1)	-0.001 [0.002]	0.006* [0.003]	-0.005* [0.003]
Equity to liabilities (t-1)	0.000*** [0.000]	0.000*** [0.000]	-0.000 [0.000]
Loan quality (t-1)	-0.002* [0.001]	-0.002 [0.001]	-0.002 [0.001]
Bank age (t-1)	0.130* [0.071]	0.246** [0.116]	0.028 [0.086]
Year Fixed Effects	Yes	Yes	Yes
Bank x Deposit-Type Fixed Effects	Yes	Yes	Yes
Observations	4,782	1,564	3,218
R-squared	0.125	0.182	0.118

Note: The dependent variable is deposit growth, measured as the first difference of the natural logarithm of bank deposits. Pandemic is an indicator variable equal to 1 in 1918 through 1919, Preferred is an indicator variable equal to 1 for preferred deposits, and Excess Death is an excess mortality-based measure of the impact of the pandemic at the county level. Standard errors are clustered by bank and deposit type, and are reported in parenthesis.

* denotes a p-value less than 0.10, ** denotes a p-value less than 0.05, *** denotes a p-value less than 0.01.

Table 5: The Effect of the 1918 Pandemic on Bank Loan Composition, 1914 to 1925

	All	Financial Center	Rural
Pandemic x Excess Death	0.019 [0.013]	-0.047 [0.079]	0.015 [0.013]
Riskier x Pandemic x Excess Death	-0.026** [0.012]	-0.052* [0.029]	-0.019 [0.013]
Total assets (t-1)	-0.099** [0.039]	-0.124* [0.065]	-0.066** [0.030]
Reserves to assets (t-1)	0.003** [0.001]	0.003 [0.004]	0.003** [0.001]
Equity to liabilities (t-1)	-0.001*** [0.000]	-0.001 [0.004]	-0.001*** [0.000]
Loan quality (t-1)	-0.000 [0.001]	0.001 [0.002]	-0.000 [0.001]
Bank age (t-1)	-0.035 [0.041]	0.050 [0.107]	-0.048 [0.036]
Year Fixed Effects	Yes	Yes	Yes
Bank x Loan-Type Fixed Effects	Yes	Yes	Yes
Observations	5,586	1,542	4,044
R-squared	0.194	0.203	0.206

Note: Note: Dependent variable is loan growth, measured as the first difference of the natural logarithm of bank loans. Pandemic is an indicator variable equal to 1 in 1918 through 1919, Riskier is an indicator variable equal to 1 if the loans fall into the group of riskier loans, and Excess Death is an excess mortality-based measure of the impact of the pandemic at the county level. Standard errors are clustered by bank and deposit type, and are reported in parenthesis.

* denotes a p-value less than 0.10, ** denotes a p-value less than 0.05, *** denotes a p-value less than 0.01.

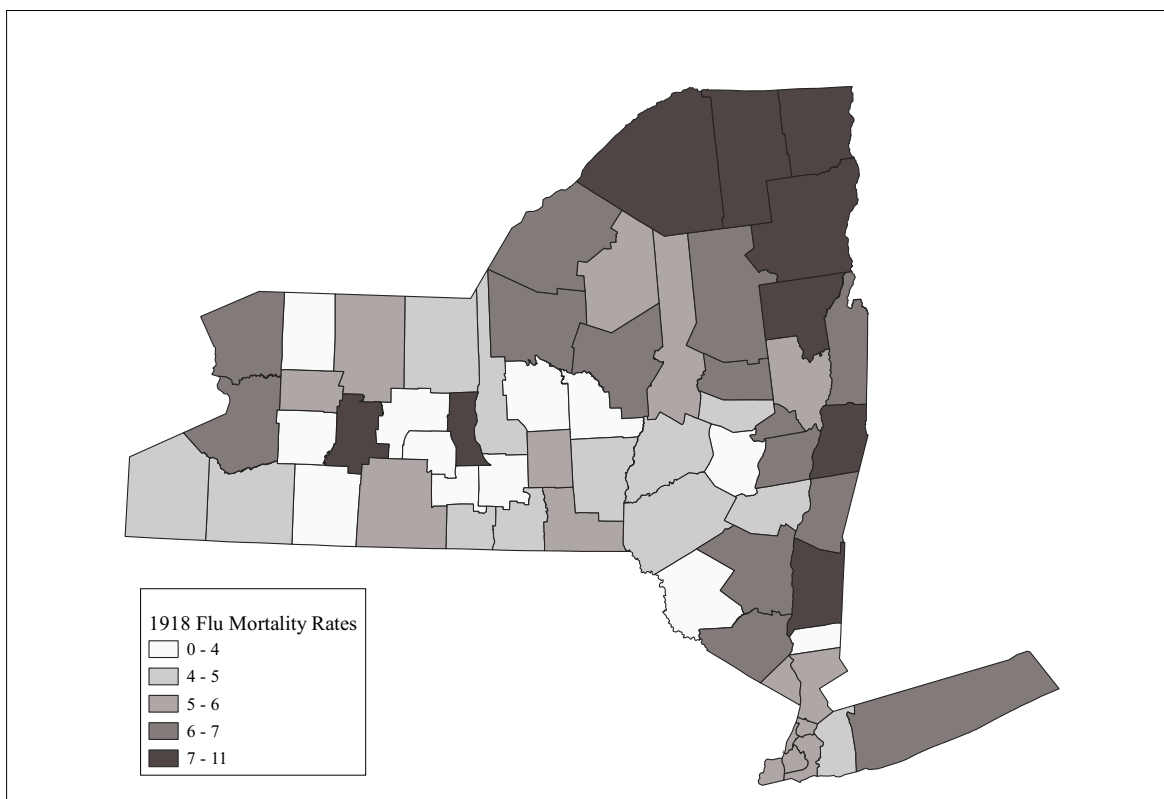


Figure 1: Pandemic Mortality Rates by County in 1918

Source: *Annual Report of the Federal Reserve Board and Annual Report of State Department of Health of New York.*

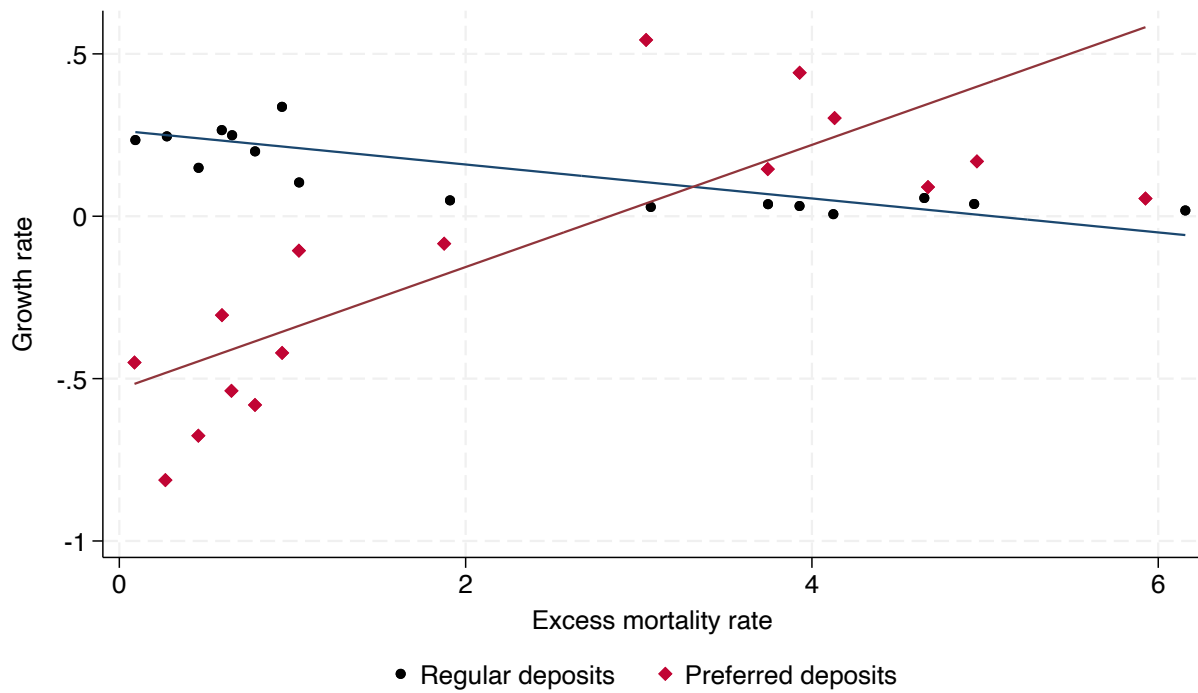


Figure 2: Empirical Relationship between Deposit Growth and Excess Mortality Rates

This figure illustrates the empirical relationship between deposit growth rates and excess mortality rates in 1918 and 1919, using a binned scatter approach. The slopes of the best fit lines for regular and preferred deposits are -0.052 (0.004) and 0.188 (0.022), respectively, with standard errors in parenthesis.

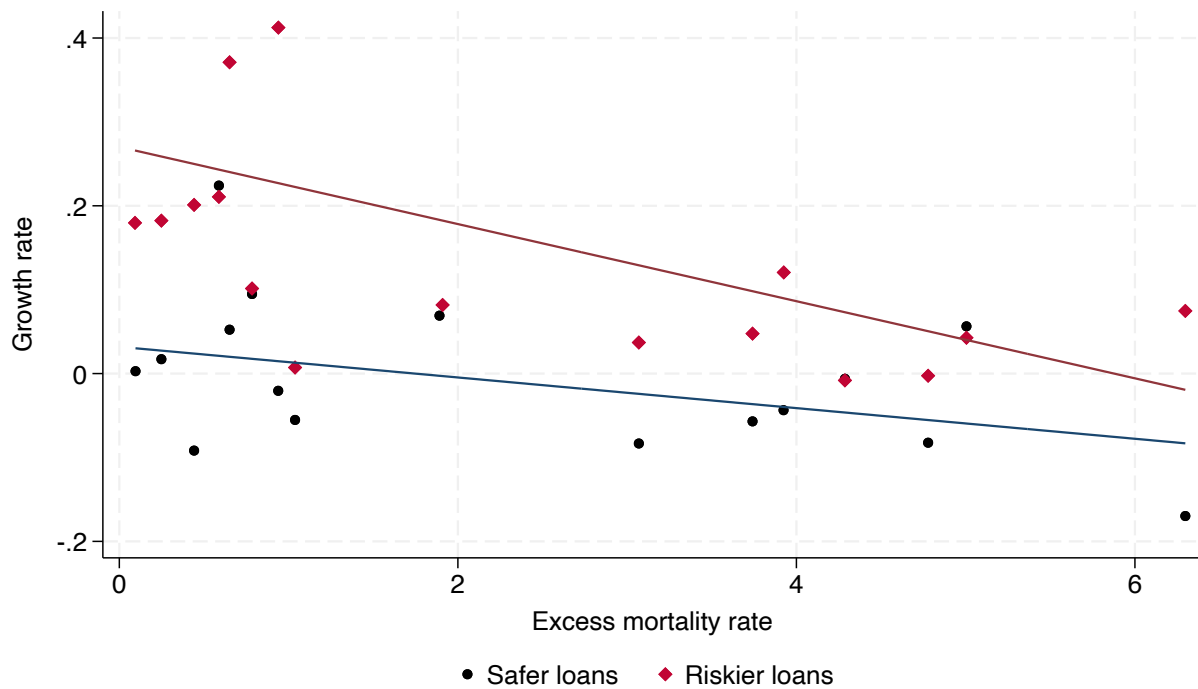


Figure 3: Empirical Relationship between Loan Growth and Excess Mortality Rates

This figure illustrates the empirical relationship between loan growth rates and excess mortality rates in 1918 and 1919, using a binned scatter approach. Safer loans are those collateralized by housing or securities, whereas riskier loans are those collateralized by other assets or not collateralized at all. The slopes of the best fit lines for safer and riskier loans are -0.018 (0.008) and -0.046 (0.007), respectively, with standard errors in parenthesis.

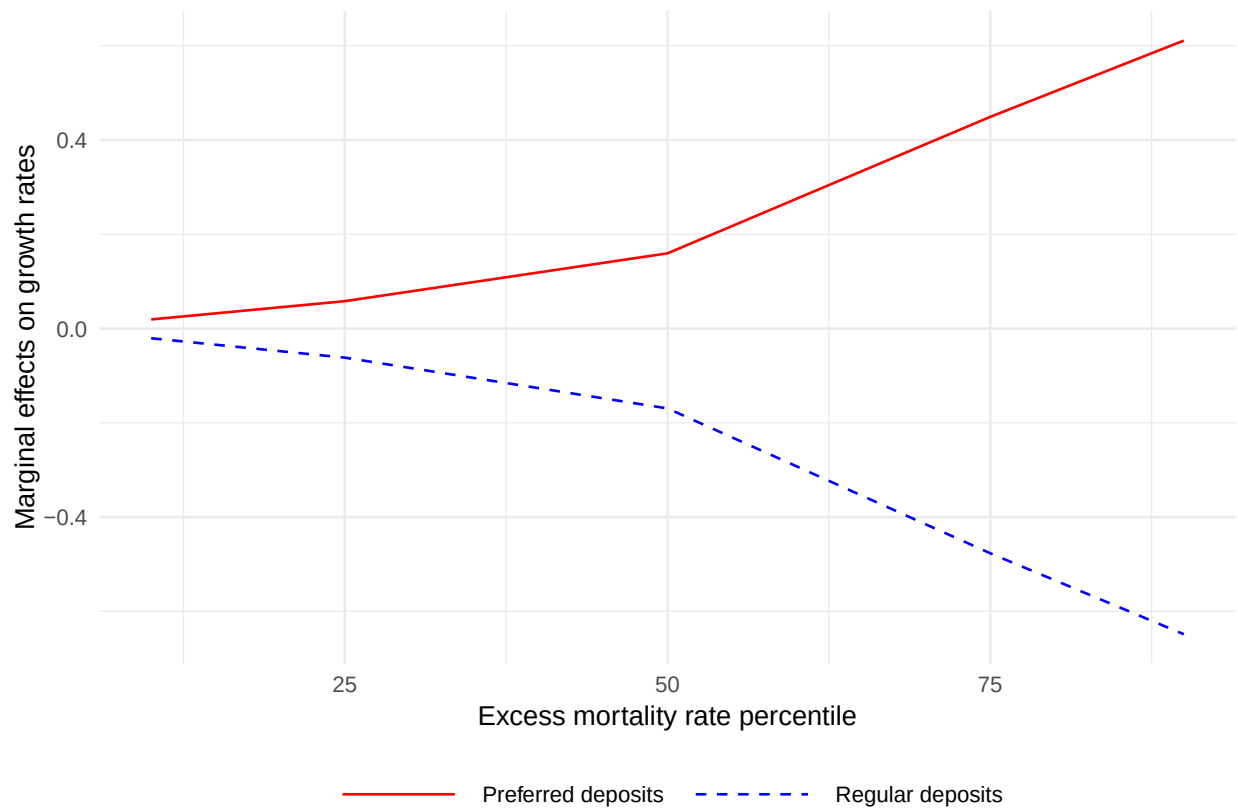


Figure 4: Marginal Effects of Excess Mortality Rates by Percentiles

Note: This figure illustrates the marginal effects of excess mortality rates on growth rates of regular deposits and preferred deposits. The x-axis is the percentile of excess mortality rates in the pandemic period, and the y-axis is the excess mortality rates multiplied by the respective coefficients in column (1) in Table 3. The blue dashed line represents marginal effects of excess mortality rates on regular deposits, and the red solid line represents marginal effects of excess mortality rates on preferred deposits. The difference in growth rates between regular and preferred deposits widens as the excess mortality rate increases.

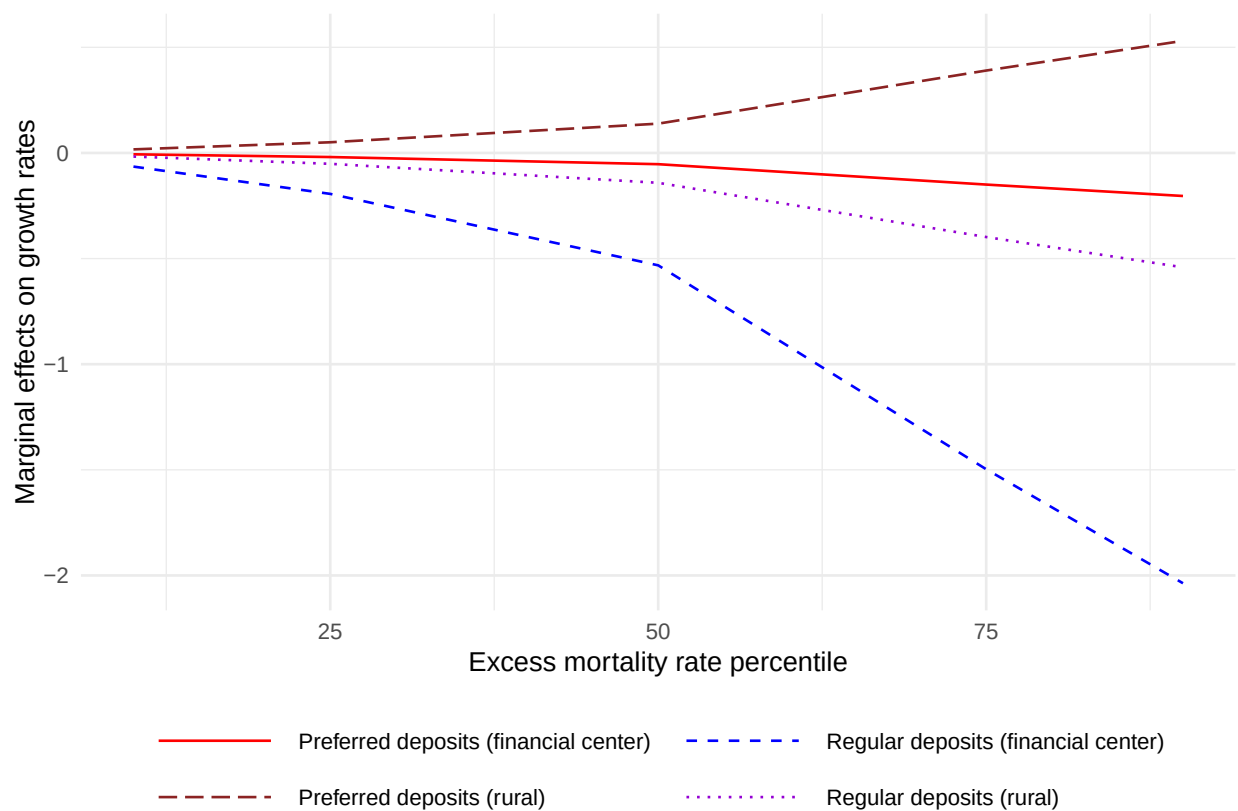


Figure 5: Marginal Effects of Excess Mortality Rates by Percentiles, by Bank Group

Note: This figure illustrates the marginal effects of excess mortality rates on growth rates of regular deposits and preferred deposits, by bank group. The x-axis is the percentile of excess mortality rates in the pandemic period, and the y-axis is the excess mortality rates multiplied by the estimated coefficients taken from columns (2) and (3) in Table 3. The brown long-dash line and the dotted line are the marginal effects of excess mortality rates on preferred and regular deposits, respectively, for rural banks. The orange solid line and the short-dash blue line are the marginal effects on preferred and regular deposits, respectively, for banks in financial centers. Deposit outflows were more severe for banks in financial centers compared to those for rural banks. For both groups, the difference in growth rates between regular and preferred deposits widens as the excess mortality rate increases.

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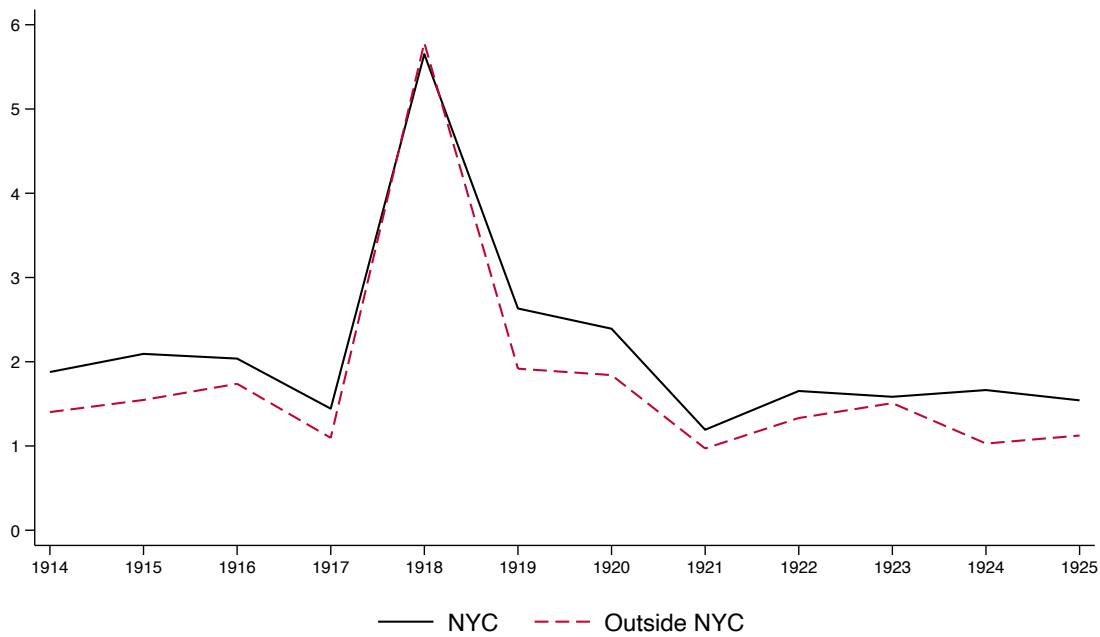
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Appendices

A New York state annual mortality rates from influenza and pneumonia

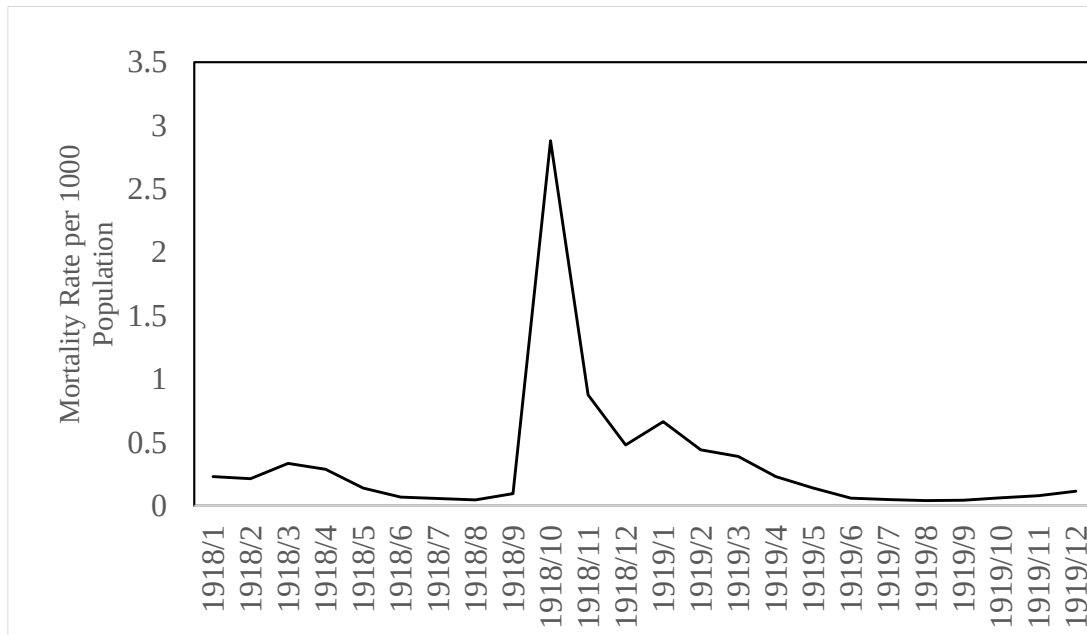
Figure A1: New York State Annual Mortality Rates from Influenza and Pneumonia per 1000 People



This figure reports NY annual mortality rates from influenza and pneumonia per 1000 people from 1914 to 1925 for counties in NYC and for those outside of NYC.

Source: *Annual Report of State Department of Health of New York*.

Figure A2: New York Monthly Mortality Rates from Influenza and Pneumonia, 1918-19



This figure report NY monthly mortality rates from influenza and pneumonia per 1000 people in 1918 and 1919.
Source: *Annual Report of the State Department of Health of New York*.

Table A1: Summary Statistics of NY County-Level Mortality Rates from Influenza and Pneumonia

	Mean	SD	Percentiles				
			10	25	50	75	90
Mortality Rates							
Excluding pandemic	1.42	0.49	0.87	1.10	1.37	1.69	2.04
Pandemic	3.69	2.15	1.55	1.87	2.74	5.36	6.67
Excess Mortality Rates							
Pandemic	2.30	2.09	0.17	0.51	1.40	3.94	5.36

Note: SD is standard deviation. Underlying data are NY county-level mortality rates where the cause of death is influenza or pneumonia. The sample period is 1914 to 1925 and the pandemic occurs in 1918 and 1919. There are 116 observations in the pandemic period and 579 observations in the complement of the pandemic period. See Appendix Section C for details on how the excess mortality rate is calculated and note that it is computed only for the pandemic period.

Source: *Annual Report of the State Department of Health of New York and authors calculations.*

B Description of balance sheet variables

Table B2: Assets and Liabilities Reported, New York State-Charter Banks

Assets		Liabilities	
Securities	Stocks and bonds, viz.: Public securities (market value) Private securities (market value)	Equity capital	Capital Surplus, including all undivided profits (market value)
Total loans	Mortgages owned Loans and discounts secured by bond and mortgage, deed or other real estate collateral Loans and discounts secured by other collateral Loans, discounts, and bills purchased not secured by collateral	Deposits	Preferred deposits Due depositors, not preferred
Interbank reserves	Due from the Federal Reserve Bank of New York less offsets Due from trust companies, banks, and bankers	Short-term borrowings	Bills payable Rediscounts
Vault reserves	Specie Legal tender notes and notes of national banks Federal reserve notes Cash Items	Other liabilities	Due to trust companies, banks, and bankers Acceptances of drafts payable at a future date or authorized by commercial letters of credit Other liabilities Add for cents
Other assets	Customers' liability on acceptances (per contra, see liabilities) Other assets Real estate owned Add for cents Overdrafts		

Source: New York State Banking Department (1914-1925).

Table **B2** displays the set of balance sheet variables of state-chartered banks and trust companies recorded in the Annual Report of the Superintendent of Banks and how we aggregate these variables for our analysis. On the asset side, regulators report a fine breakdown of bank reserves which we group into two categories: vault reserves and interbank reserves. The vault reserves were composed of “species”, “legal tender notes and notes of national banks”, “federal reserve notes”, and “cash items”. This collection of items are currency or currency-like instruments that are held at the bank. Interbank reserves, or reserves that a bank has deposited in other banks or at a Federal Reserve Bank, is the sum of ‘due from trust companies, banks and bankers’ and ‘due from the Federal Reserve Bank.’ The structure of the banking system in the early 1900s was a tiered system by which smaller banks placed reserves at larger banks located in reserve cities ([Anderson, Padrik and Wang, 2019](#)). This structure was formalized in that banks that were not members of the Federal Reserve (nonmember banks) were required to meet their reserve requirements by holding

deposits in these reserve-city banks. Banks that were members of the Federal Reserve (member banks) met their reserve requirements by placing cash at the Federal Reserve.²⁴ However, these member banks also placed cash at reserve city banks, because these interbank deposits, which were generally considered safe assets, earned interest, whereas deposits held at the Federal Reserve did not.

New York state regulators also collected detailed information on the loans made by banks, placing them into four categories. The first category is ‘mortgages owned,’ which are real estate loans. State banks were allowed to make mortgage loans against farmland within one hundred miles of their location (or headquarters) with several restrictions. The second category is ‘loans and discounts secured by bond and mortgage, deed, and other real estate collateral,’ which are loans collateralized by mortgage security and deeds.²⁵ The third category is ‘loans and discounts secured by other collateral,’ which are loans collateralized by anything except for real estate security, Liberty Bonds, and securities. The fourth and final category is ‘loans, discounts, and bills purchased not secured by collateral,’ which are promissory notes. We aggregate the first two categories as Safer loans and the second two categories as Riskier loans. We also create a measure of the quality of a bank’s loan that is equal to the ratio of secured loans over total loans, a variable that is used as a bank-level control in the analysis.

The liability side of the balance sheets was mainly composed of capital, deposits, and short-term borrowing, where capital and deposits comprise over 90 percent of total liabilities. We aggregate capital and equity into one measure called equity capital, which is equal to the sum of ‘capital’ and ‘surplus, including all undivided profits.’ We calculate total deposits as the sum of ‘preferred deposits’ and ‘due depositors, not preferred,’ and define short-term borrowing as the sum of ‘rediscounts’ and ‘bills payable.’ Rediscounts are loans sold with recourse, and bills payable are promissory notes issued by the bank. During our sample period, non-member banks primarily borrowed from correspondent banks located near towns. These correspondent banks were national banks that were members of the Federal Reserve.²⁶ On the contrary, member banks borrowed

²⁴Only Federal Reserve System members were eligible to place cash reserves at their regional Federal Reserve Bank.

²⁵A deed was taken as a mortgage but not an absolute transfer of ownership.

²⁶The Federal Reserve Act made it compulsory for national banks to become members of the Federal Reserve

directly from Federal Reserve Banks.²⁷

System, while it made it voluntary for state banks to become members.

²⁷For more information, see [Carlson and Wheelock \(2018\)](#) and [Anderson, Erol and Ordoñez \(2022\)](#).

C Excess mortality rate construction details

In this section, we provide details on how the excess mortality rate is calculated. This is done by using the annual data on mortality statistics for each county. The mortality measure, denoted $r_{i,t}$, is the ratio of deaths from influenza and pneumonia over population in county i in year t . We predict this measure using the natural log of the county's population (denoted $\ln \text{pop}_{i,t}$), with fixed effects for each county. Formally, the specification is:

$$r_{i,t} = \alpha_0 + \alpha_1 \ln \text{pop}_{i,t} + \eta_i + \varepsilon_{i,t}, \quad (\text{C1})$$

where η denote county fixed effects and ε is a error term.

Importantly, we estimate this model using annual data from 1914 to 1925, with 1918 and 1919 excluded. As a result, this model captures the usual mortality rates in a county, excluding the direct effects of the 1918 pandemic. The estimated coefficients from the regression are reported in Appendix Table C3. These coefficients are then used to predict mortality rates for each county in 1918 and 1919, denoted $\hat{r}_{i,t}$. Finally, a measure of excess mortality rate for each county in 1918 and 1919 is calculated as the difference between the actual mortality rates and the predicted values,

$$x_{i,t} = r_{i,t} - \hat{r}_{i,t}. \quad (\text{C2})$$

This excess mortality measure is then merged with the annual bank-level data.

Table C3: Estimated coefficients of the mortality model

Log of population	-1.181*** (0.272)
Constant	15.905*** (3.309)
R-squared	0.27
Observations	579

This table reports the estimated coefficients from a model predicting mortality rates from influenza and pneumonia. Annual data from 1914 to 1925, but excluding the pandemic years of 1918 and 1919, are used to estimated the model. County-level fixed effects are included in the regression but are not reported.

* denotes a p-value less than 0.10, ** denotes a p-value less than 0.05, *** denotes a p-value less than 0.01.

D Federal reserve system membership

A major change to the banking system in 1910s was the establishment of the Federal Reserve System. It was created by the passage of the Federal Reserve Act of 1913 and went into operation on November 16, 1915. National-charter banks, which were regulated by the Office of the Comptroller of the Currency (OCC), were required to become members of their local Federal Reserve Bank. State-charter banks, which were regulated by their state's banking regulator, had the option of becoming members.

A main value of being a member was gaining direct access to the discount window. During this period, the Federal Reserve conducted monetary policy exclusively through discount window lending (Chabot, 2017).²⁸ The Fed's founders had not considered monetary policy in the modern sense, so liquidity provision by the Federal Reserve was passive. Each Federal Reserve Bank set a discount rate, but Federal Reserve credit was provided at the initiative of banks who came to the discount window to borrow reserves (Wheelock, 2013). In contrast to later years, during the early years of the Federal Reserve System, banks did not suffer from stigma when they accessed the discount window (Gorton and Metrick, 2013).

State-chartered banks were eligible for membership in the Federal Reserve System if they met certain criteria. Upon becoming a member, state-charter banks became subject to federal banking regulations and supervision, which were sometimes more strict than those of the relevant state banking regulator. At the same time, a state-chartered bank could still enjoy the benefit of the Federal Reserve's discount window by establishing a correspondent relationship with a bank that was a member (Anderson et al., 2018). Another cost was that banks which were members of the Federal Reserve System were required to hold their required reserves at a Federal Reserve Bank where these reserves did not earn interest. In contrast, non-members held their required reserves at other banks and earned interest.

Given the trade-offs of membership, few state-charter banks joined the Federal Reserve System when it was first established. In 1917, the Federal Reserve introduced major reforms to the membership process for state-charter banks to make membership more appealing. As a result, there was an uptick in the number of state-charter banks becoming members of Federal Reserve System in 1917 and through the end of the sample period. Nevertheless, many state-charter banks still did not join the Federal Reserve System. In Appendix Figure D3, the number of NY state-charter banks which are members of the Federal Reserve System is illustrated alongside the equivalent nationwide number. This comparison shows that the evolution in the number of state-charter banks becoming members of the Federal Reserve System in NY mirrors what has happening in the U.S.

²⁸Until 1922, there were no open market operations to conduct monetary policy. Open market purchases were conducted only to support Treasury issuance or raise revenue of the Federal Reserve Banks.

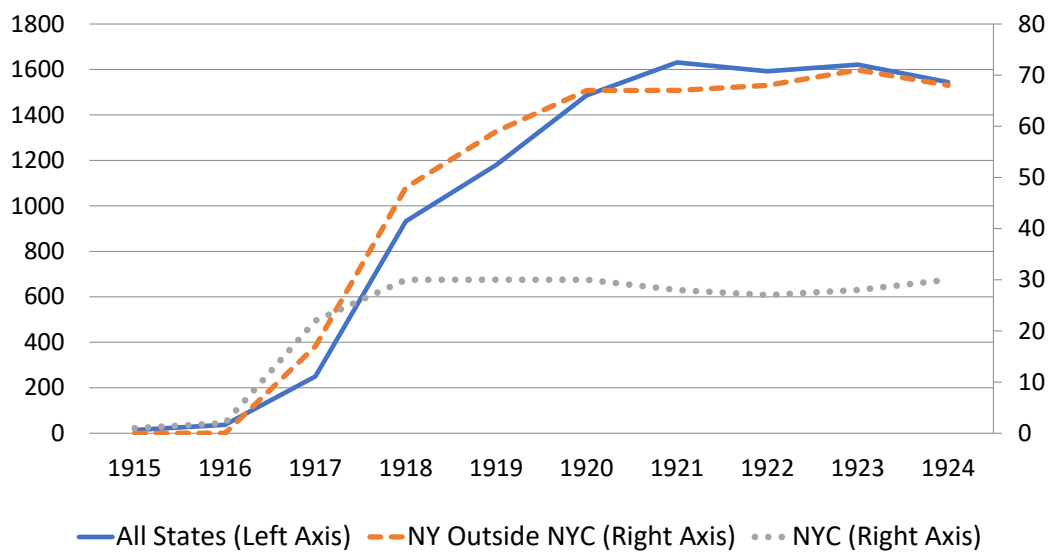


Figure D3: Number of State-Charter Bank that are Federal Reserve System Members (1915-1924)
 This figure displays the number of state-charter banks which chose to become members of the Federal Reserve System from 1915 to 1924 across the US and in NY. Information on bank membership is provided in the *Annual Report of the Federal Reserve Board*.

E Additional tables

Table E4: Depositors' Reactions to the Severity of Pandemic, 1916 to 1919

	All	Financial Center	Rural
Pandemic x Excess Death	-0.115*** [0.027]	-0.411*** [0.139]	-0.101*** [0.026]
Preferred x Pandemic x Excess Death	0.243*** [0.025]	0.341*** [0.054]	0.209*** [0.027]
Total assets (t-1)	-0.837*** [0.139]	-0.353*** [0.071]	-1.185*** [0.196]
Reserves to assets (t-1)	0.009*** [0.003]	0.003 [0.003]	0.008** [0.004]
Equity to liabilities (t-1)	0.015*** [0.004]	0.000*** [0.000]	0.012*** [0.004]
Loan quality (t-1)	0.001 [0.001]	-0.001 [0.001]	0.002 [0.001]
Bank age (t-1)	0.636* [0.341]	0.258** [0.105]	0.763** [0.318]
Year Fixed Effects	Yes	Yes	Yes
Bank x Deposit-Type Fixed Effects	Yes	Yes	Yes
Observations	1,824	1,391	1,186
R-squared	0.317	0.222	0.322

Note: This is the similar to Table 3 except the regression is estimated over a shorter sample period, 1916 to 1919.

* denotes a p-value less than 0.10, ** denotes a p-value less than 0.05, *** denotes a p-value less than 0.01.

Table E5: The Effect of the 1918 Pandemic on Bank Loan Composition, 1916-1919

	All	Financial Center	Rural
Pandemic x Excess Death	0.035** [0.014]	-0.046 [0.084]	0.033** [0.014]
Riskier x Pandemic x Excess Death	-0.037*** [0.013]	-0.055* [0.029]	-0.030** [0.014]
Total assets (t-1)	-0.326*** [0.119]	-0.142* [0.075]	-0.135 [0.093]
Reserves to assets (t-1)	0.007*** [0.002]	0.003 [0.004]	0.007*** [0.003]
Equity to liabilities (t-1)	-0.003 [0.003]	-0.001 [0.004]	0.001 [0.002]
Loan quality (t-1)	-0.000 [0.001]	0.001 [0.002]	-0.001 [0.002]
Bank age (t-1)	0.045 [0.189]	0.057 [0.112]	-0.221 [0.190]
Year Fixed Effects	Yes	Yes	Yes
Bank x Loan-Type Fixed Effects	Yes	Yes	Yes
Observations	1,930	1,338	1,312
R-squared	0.323	0.203	0.288

Note: This is the similar to Table 5 except the regression is estimated over a shorter sample period, 1916 to 1919.

* denotes a p-value less than 0.10, ** denotes a p-value less than 0.05, *** denotes a p-value less than 0.01.